Research Topic: Leveraging business analytics for strategic decision-making: A survey on data-driven management practices.



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Submitted to: Dr. Sumesh DADWAL

Submitted By: Sudar Ramakrishnan

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Abstract

The rise in use of business analytics has changed the manner in which organisations make strategic decisions in highly complex and uncertain environments. The research evaluates the role of data quality, analytics capability, and managerial support in the effectiveness of strategic decisions in data-driven organisations. Quantitative research design was chosen based on the "Information Processing Theory and the Resource-Based View" because it involves the cross-sectional survey of 111 professionals in various industries. The SPSS was used to test the hypotheses proposed by running descriptive, correlation, and regression analysis. The results indicate that the quality of data, analytics capability, managerial support and strategic decision-making effectiveness have strong and statistically significant relationships. Also, the presence of managerial support and analytics capability was identified to have a substantial impact on the adoption of business analytics. The integrated model justifies a significant level of variance in the performance of decision making, where the technological, data and leadership resources are complementary. The research study has a contribution to the available literature through the empirical validation of an integrated framework of analytics-based strategy and practical contribution to organisations that require the improvement of the quality of decisions based on the balanced investments in the improvements of data governance, analytical capabilities, and supportive of leadership.

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Chapter 1: Introduction

1.1 Background to the Study

The increasing use of data in strategic decision-making has created new ways of organizations reacting to uncertainty, enhancing performance, and remaining competitive. With the increased sophistication of analytics tools, the strategic decision-making process has also evolved less into the use of intuition-based approaches and more towards evidence-based ones. According to Klievink et al. (2017), this change is also due to the fact that business analytics (BA) allows combining various types of data: structured, unstructured, internal, and external into information that supports decisions. Instead of concentrating on the mass of the collected data, the organizations are becoming more engaged in the reading of the said data into meaningful insights that can assist the executives with a more accurate evaluation of the available alternatives and allowing them to react in the quickest possible way to the arising challenges (Klievink et al., 2017). This development underscores the reason why lots of companies today consider analytics as a business asset, and not a technical operation.

In this transition, data quality is one of the most significant factors in the effective functioning of decisions. When the data are true, full, and are provided in time, then managers are able to detect the patterns at a better rate and develop an evidence-based decision-making process. According to Hurbean et al. (2024), organizations that are determined to ensure the accuracy of their data have more likely to have positive relationships between the implementation of analytics and data quality of decisions, since the credibility of inputs enhances the accuracy of insights. Their work also indicates that data quality ensures that operations run well and managerial results improve partly due to the fact that managers have more confidence in the information they use in making decisions. Conversely, low quality data tends to generate uncertainty, misrepresents results of analytic processes and diminishes the value of analytics instruments. Therefore, data quality is a technical need as well as a starting point of the strategic rationale in data-oriented settings (Hurbean et al., 2024).

Together with the quality of data, the analytics capability of an organization determines the accuracy and timeliness of strategic decisions. The capability of analytics typically portrays the cumulative power of analytical capabilities, technological assets and infrastructural frameworks that transform raw information to actionable intelligence. According to Pedota (2023), the real worth of analytics is not merely processing data, but making and sharing insights in such a manner that they contribute to learning the strategic aspects. This is

particularly crucial in dynamic situations where the responsiveness might be deterred by delays in analysis. Practical examples support this fact: Transport for London, to name an example, applies analytics to real-time management of transport flows, and it shows that developed analytical abilities can improve strategic flexibility (Kandt & Batty, 2021). These are examples that explain why analytics capability is the key element in organizations that are likely to make informed decisions at a fast pace.

The other significant pillar that contributes to the adoption and efficacy of data-driven practices is the managerial support. The leaders supporting analytics efforts indicate the availability of sufficient resources, employee credibility with the results of analytics, and adoption of data-based decision-making as a standard practice. According to Szukits and Moricz (2024), the high-level managerial support enhances the impressions of data quality and strengthens an analytics-oriented culture. They however, also warn that managerial support is not always a guarantee of truly data-driven decision-making, when the rights to make decisions are over-centralized. They note that balanced leadership is necessary, which fosters teamwork and allows the teams to make sense of analytical findings instead of focusing only on receiving top-down instructions (Szukits & Moricz, 2024). This point is a hint of how complicated the role of leaders in analytics adoption can be.

Collective experience serves as the data quality, analytics capability, and managerial support, which in effect impacts the overall strategic decision making performance. According to Duan et al. (2020), organizations that implement analytics tools along with a disposition focused on data are likely to have more successful innovation results and better performance. Equally, Liang et al. (2022) posit that the strategy of firms that combine analytical systems with balancing the opportunities found in the search and exploiting the strength available in the present is better strategic and achieves the best outcome. This point is bolstered by Khaddam (2024) who demonstrates that ambidexterity based on analytics can empower the future-related decision-making and operational enhancements. Combined, these views indicate that strategic decisions quality is hardly influenced by a factor alone; instead, it is a result of a combination of good-quality data, analytical skills, and leadership that promotes the thinking process based on data.

1.2 Research Aim

To examine how the use of business analytics enhances strategic decision-making effectiveness in organizations by assessing the role of data quality, analytics capability, and managerial

support.

1.3 Research Objectives

1. To evaluate the relationship between data quality and strategic decision-making effectiveness in data-driven organizations.
2. To assess how organizational analytics capability influences the accuracy and timeliness of strategic decisions.
3. To determine the effect of managerial support for data-driven practices on the adoption and effectiveness of business analytics in strategic decisions.
4. To examine the collective impact of data quality, analytics capability, and managerial support on overall strategic decision-making performance.

1.4 Research Questions

1. What is the relationship between data quality and strategic decision-making effectiveness in data-driven organizations?
2. In what ways organizational analytics capability influences the accuracy and timeliness of strategic decisions?
3. What is the effect of managerial support for data-driven practices on the adoption and effectiveness of business analytics in strategic decisions?
4. What is the collective impact of data quality, analytics capability, and managerial support on overall strategic decision-making performance?

1.5 Scope of study

The present research is concerned with comprehending the influence of data quality, organizational analytics capacity, and managerial assistance as the three factors that define the efficiency of strategic decisions making in data-driven organizations. The sample is narrowed down to mid- and large-scale companies that already incorporated business analytics in their planning and performance review procedures and this guarantees that the respondents have the experience of using analytics-based decision environments. Focusing on companies where analytics usage is established implies that the study can better describe the interaction of these three factors in actual operation environments, as opposed to evaluating hypothetical or early adoption environments (Hurbean et al., 2024). The study also explores how much these variables have an impact on the accuracy and timeliness of strategic decisions, which indicates the growing importance of empirical evidence in the contemporary managerial practice

(Klievink et al., 2017). The research fails to examine industry-specific and long-term causal effects because it collects data at one time. Rather, it focuses on the need to find patterns and relationship processes that can be used to explain why certain organizations perform better on decision performance when applying analytics-driven processes (Bryman, 2016).

1.6 Research methodology

The methodological scheme is based on a quantitative design, which makes it possible to organize the investigation of the relations between the data quality, analytics capability, managerial support, and the effectiveness of strategic decision-making. Quantitative designs are optimal in the research aimed at measuring the interactions among measurable constructs and determining the statistical significance of them (Creswell & Creswell, 2018). In this respect, the study will utilize a survey cross-section method because it will collect data at a single point in time about organizations that are actively using analytics in their strategic processes. The cross-sectional designs are especially suitable when it is necessary to investigate the correlation between organizational variables without long-term observation (Bryman, 2016).

The information will be gathered by means of a structured, closed-ended questionnaire on the five-point Likert scale to make sure that all respondents will be tested equally. The instrument is based on standard measurement scales that were derived out of previous research to enhance reliability and construct validity. A piloting test will be carried out with a small sample of professionals to clarify the points, based on the practical advice of Connelly (2016). The purposive sampling is employed to reach managers and analytics specialists who are directly involved in the strategic decision processes because of the fact that this group has the knowledge that is applicable to the aims and objectives of the study (Saunders et al., 2009). The sample size will be 100-150 respondents, which is adequate to give statistical power to conduct regression analysis.

1.7 Structure of the study

The study is composed of 6 major sections as the figure below depict. The study also reflect on the journey of the researcher throughout the module and particularly the process of developing the study.

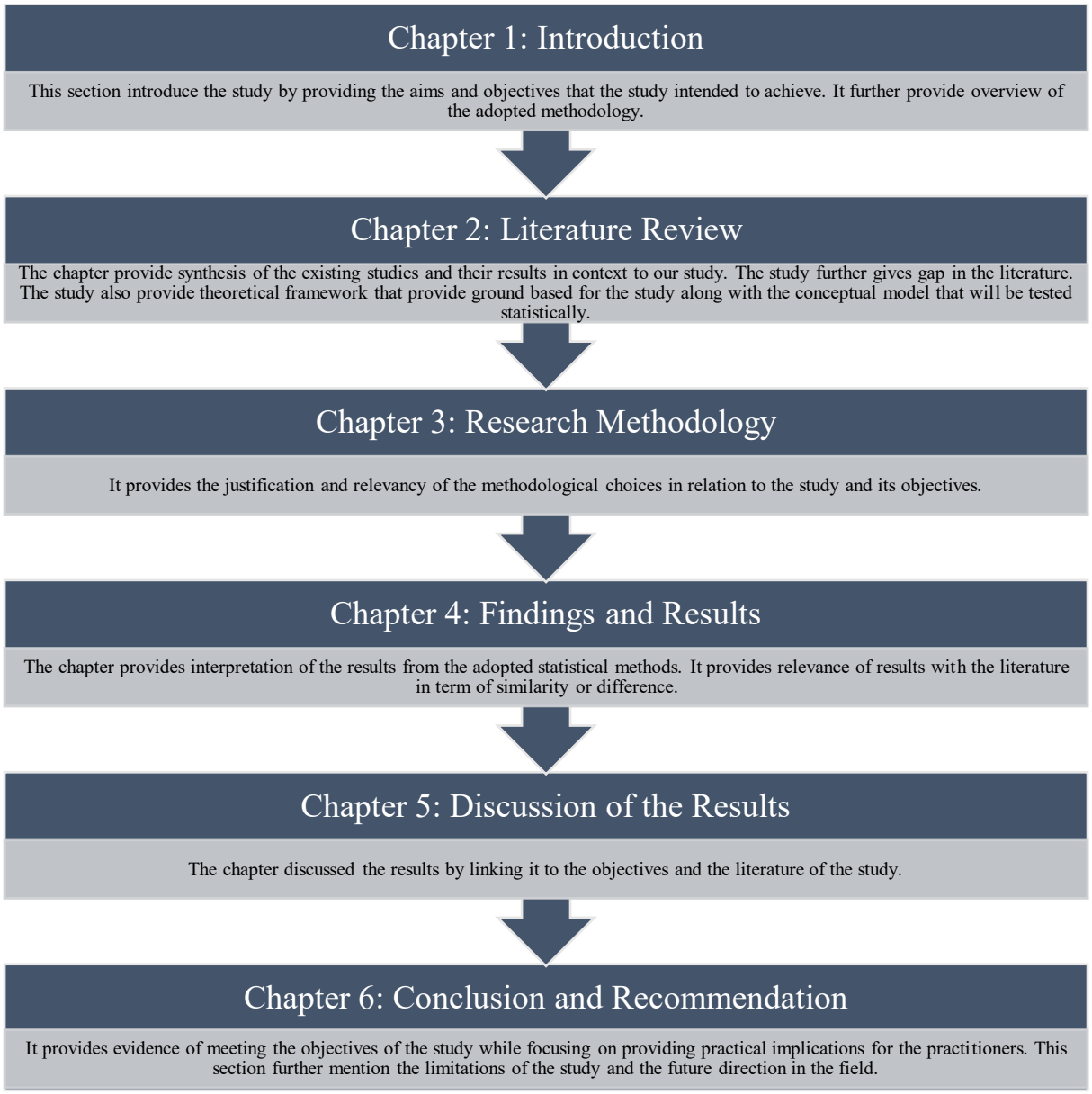


Figure 1: Structure of the study

Chapter 2: Literature Review

2.1 Introduction

The chapter overviews the theoretical foundations that are important in the study of strategic decision-making in data-driven organisations. It is a synthesis of the research done on the quality of the data, the ability to conduct analytics, and support managers to determine their applicability in the decision-making effectiveness. Through the analysis of the interaction of these factors in the context of analytics-infused settings, the chapter identifies significant developments, contradictions, and gaps in the literature. Another benefit of this review is its conceptual foundation of the proposed framework because it links the existing theories to the current empirical evidence.

2.2 Strategic Decision-Making in Data-Driven Organizations

The process of strategic decision making in organizations has experienced a dramatic change as companies move away with the use of intuition in making decisions to methods that are based on data and analytical data. In the past, the management judgement was based on experience and qualitative-based judgement which in many instances restricted the potential to predict risks or react to quick market trends. As more and more digital data and tools of analysis have become accessible, this has changed. Practices that are transparent, defensible, and informed by evidence are growing as a natural response of contemporary strategic environments, which heightens the need to move toward data-driven practices that help organisations be increasingly responsive (Klievink et al., 2017). The development does not represent the denial of intuition but its fine-tuning, the assistance of human intuition by systematic evidence to enhance its accuracy and strategic consistency.

Business analytics (BA) has been at the centre of this revolution, with business analytics becoming a major facilitator of contemporary organisational strategy. BA eases the process of consolidating both structured and unstructured data enabling firms to create real-time insights and reveal subtle patterns that would not be immediately apparent on a traditional analysis. Further technologies, particularly cloud technologies, machine learning models, and AI-based algorithms, have enabled working with large data volumes and making the forecasting, risk assessment, and operational optimisation more effective (Laudon & Laudon, 2020). Nevertheless, BA is still an emerging area in most organisations even with its increasing influence. According to Kunc and O'Brien (2019), analytics tend to be implemented without a well-designed framework, which leads to the fragmentation of practices and suppresses its

strategic potential. According to their analysis, BA should be entirely assimilated with the accepted decision-support processes in order to be significant in influence on strategic decisions as opposed to being a standalone technical purpose.

Data-driven culture has become a key requirement in the successful adoption of analytics. These cultures put evidence and not hierarchy first, promote experimentation, and promote the continuous application of analytics across all levels of the organisation. Based on resource-based views, Aydiner et al. (2019) reveal that the adoption of analytics leads to the better performance of business processes, which consequently leads to better organisational performance- indicating that analytics should be incorporated into the everyday business workflow, and not implemented selectively. Evidence-based cultures enable managers to break the consensus and minimise the excessive reliance on gut feeling, developing a more balanced and disciplined approach to strategic choice. Simultaneously, a data-driven culture should be cultivated not only with organisational dedication but also managed sponsorship because employees need to have confidence in its analytical products and feel encouraged to use them on a regular basis.

Although analytics-based decision-making has great potential, there are a number of issues to consider, which the organisations should manage. The advantages of BA agility, competitive advantage, and quality of information have been severally proven in empirical research. Specifically, as an example, Zamil et al. (2023) indicate that organisations with more developed analytics practices have greater competitive advantages in particular when analytics is in line with strategic goals. Likewise, the article by Bhisikar (2024) in the transportation domain shows that strategic business analytics enhances the connection between business strategies and competitive advantage, which proves the strategic capabilities of analytics integration. Nevertheless, there are challenges associated with such opportunities, such as privacy issues, problems with data integration, the lack of transparency in the algorithms, and the necessity of niche skills. Research on the dynamic environment also insinuates that BA capabilities affect the performance of firms in part by enhancing organisational agility and innovation capacity, though this is moderated by industry volatility (Khan et al., 2024). These observations underscore the fact that the achievement of analytics is not automatic and uniform but relies on situational factors, organisational preparedness, and the capacity to handle technological and cultural complexities.

2.3 Data Quality and Its Role in Strategic Decision-Making

The quality of data has emerged as a core issue in the quest by organisations to make sound strategic decisions in the ever-increasing data-driven world. With the creation of digital systems that produce enormous volumes of data, capability in differentiating trustworthy data and inaccurate or unfinished data is what will ensure managerial decisions are based on sound insights and trends dependable on falsified or flawed information. There are a few fundamental dimensions that are generally seen as data quality and they include accuracy, completeness, reliability, and timeliness, so that the usefulness of information is influenced by these dimensions. Accuracy eliminates misrepresentation in the analytical models, completeness guarantees the basis of decisions is a complete as opposed to a partial picture of the business environment, reliability ensures uniformity of sources/periods and timeliness ensures that the information represents the current state of the business (Khong et al., 2023). A failure of any of these factors will cause organisations to make decisions that are not in line with the realities on the ground and this results in performance failures or strategic drift.

The effect of the quality of data on the outcomes of analytical results has already become one of the well-known in the literature. Research on the big data analytics always reveals that even the advanced tools of analytics are not able to correct poor quality of inputs. Li et al. (2023) show that big data analytics can be used to improve the quality of organisational decisions considerably, but this depends on the underlying capabilities of analytics, i.e. analytical tools can only generate meaningful results when there is reliable data. In comparison, inconsistent or erroneous datasets introduce noise in the process of analytics decreasing accuracy of forecasting and constraining the usefulness of decision-support models. This brings out a larger ingredient data quality works as the platform under which analytical intelligence is developed and without it, even advanced analytics cannot be made effective in terms of their strategic impact.

The quality of data also leads to organisational learning and building evidence-based practices. Companies which focus on quality data are more likely to develop cultures of open knowledge sharing in which employees are more likely to use documented evidence instead of assumptions. A study by Ghasemaghaei (2019) demonstrates that data analytics enhances internal knowledge exchange that mediates its effect on the quality of decisions. In spite of the fact that the knowledge sharing does not necessarily lead to better decision outcomes, its value is more likely to be high in case of the combination with high-quality information and high

analytics competence. It implies that data quality enhances the learning processes by giving the employees reliable information that can be disseminated, perceived and implemented in a unified manner. In companies with strong data governance, managers become more convinced in the products of analytical processes, which strengthens the transition to evidence-based decisions and minimises the use of intuition as the only factor (Hurbean et al., 2024).

Meanwhile, the attitude towards data inconsistencies displays a more involved image. Although the high-quality data is typically considered as a strategic asset in firms, the inconsistencies of large datasets are commonly inevitable in an environment that is marked by swift change or decentralisation information flows. Ghasemaghaei and Calic (2019) demonstrate such complexity by providing the example that the use of big data can enhance some dimensions of data quality, including representational and contextual quality, but can also erode the intrinsic quality because of noise or untested inputs. Their results emphasize a paradox: the increase in the number of data sources increases the analytical possibilities but can also seriously damage the fundamental quality dimensions, when the mechanisms of governance are weak. Such discrepancies in the strategic contexts may cause indecisiveness among decision-makers, low degree of diagnostic accuracy, and differences in the interpretation of identical information. It is therefore possible that organisations both gain and lose the advantages and the disadvantages of large-scale data adoption, depending upon the strength of their quality assurance procedures.

2.4 Organizational Analytics Capability

Organizational analytics capability can be defined as a firm's ability to transform raw, complex and sometimes unstructured data into valuable insights that can be used to make informed decisions. This capability is normally based on three interdependent elements namely, technical resources, human expertise, and infrastructural systems. Technical resources refer to analytical tools, data management platforms or statistical or machine learning applications which allows organisations to process large volumes of data. Human expertise entails the analytical capacity, the knowledge of the domain and the capability of the employees to interpret data in a manner that is conciliated with the strategic objectives. The infrastructural systems imply the organisational routines, data governance structures, and the business intelligence framework, which can facilitate the on-demand flow and integration of data (Mishra et al., 2019). When these elements work together, then they form the groundwork required to draw quality insights instead of bits of information.

The efficiency of these elements is in the possibility of converting data into operational intelligence. Technical systems by themselves do not make value, unless backed by individuals, that are able to contextualise outputs and infrastructures that can guarantee consistent data flows. Research always shows that analytics capability is strategically relevant when the implementation of technology is synchronized to the capabilities of human beings. Mishra et al. (2019) demonstrate that big data and predictive analytics only diffuse within organisations when IT implementation and HR functioning combine, eventually affecting the organisational performance. Their results imply that despite the sophisticated technology, poor human or infrastructural support restricts the ability of the organisation to make meaningful interpretation of the data. However, as compared to where companies combine expertise of analysts with sound IT systems, analytical routines are within the decision-making processes and as such raw data transform into insights that inform strategic and operational decisions.

In the working environment, the capability of analytics assists organisations in streamlining the efficiency of their processes, optimising the allocation of their resources, and responding swiftly to new issues. Indicatively, Ilmudeen (2021) concludes that big data analytics capability reinforces business intelligence infrastructure, which in turn improves business operational and marketing performance. This association underlines the role of infrastructural factors like data warehouses, reporting, or foresightful monitoring instruments in closing the technical and human aspect of analytics capability. The absence of such infrastructures makes analytical efforts disjointed and hence contributing less to operational performance. The results also highlight an effective discrepancy: whereas analytics capability has a direct positive effect on certain aspects of performance, its actualisation is possible only in the case of a strong analytical infrastructure, which is an indication of the stratification of data-driven operations.

Strategically analytic capacity helps in sensing the market, innovation and long term flexibility. Organisations that have high sensing capacity are in a better position to learn the changing needs of the market, to see opportunities as well as to glimpsing on the presented disruption. According to Fosso Wamba et al. (2024), strategic outcomes are reinforced by analytics-enabled sensing and analytics culture of an organisation. Comparing the two aspects allows them to conclude that the capability of analytics can influence performance directly, and it indirectly influences organisational behaviour, as it creates the culture of evidence-based reasoning. Their findings suggest the mediation role of the analytics culture, which implies that even the most developed technical and human resources may need positive cultural norms that

can shape the strategic performance to the maximum. In comparison, organisations experiencing poor analytics cultures tend to not realise value of their analytical resources due to the fact that decision making is done more on the gut than on the evidence available.

Comparative research also demonstrates the performance difference between organisations that have good and poor analytics. Companies that have well-developed analytics systems and cultures of data-driven companies always have better strategic and operational performance. The analysis of the MSMEs conducted by Wong and Ngai (2025) proves this difference quite vividly as the abilities to sense and analyse the data are important contributors to the improvement of operation, and having the culture of data-driven processes enhances this process. These findings indicate that the poor analytics capacity, which can be caused by the limited tools, a relative lack of expertise, or insufficient cultural support, leads to the inability of an organisation to turn data into decisions, slowing down response times and making less accurate strategic decisions. Meanwhile, companies that have high analytics are also lucky to experience quicker learning processes, more accurate forecasts and are more innovative.

2.5 Managerial Support for Data-Driven Practices

Leadership is a core factor in the manner in which organisations embrace and incorporate analytics in their daily decision-making. Most industries have seen an increase in investments in high-order analytical devices, which have been surpassed by managerial readiness or capability of effectively utilizing them. This disparity in the banking industry, which has seen large outlays in big data, predictive analytics, and machine learning, has yet to be embraced by the senior leadership, is the point highlighted by Smith (2025). The study revealed that managers are deterred by weak executive sponsorship and lack of confidence over the usefulness of advanced analytics in order to integrate data-driven practices in strategic decision making. This comparison of technology readiness versus management reluctance indicates that this leadership approval is not merely useful but essential to analytics adoption.

Managerial attitudes also play another role in the culture of organisational analytics, either turning data into a regular decision-making input or ensuring it is a technical role that does not affect strategy. Indeed, Chaudhuri et al. (2024) demonstrate how such a favourable managerial mentality promotes the development of a culture of data that helps not only to reinforce innovation but also a culture of process improvement. Their results underline the role of leaders who are modelling sources of analytical thinking, such as soliciting evidence, promoting experimentation, or incentivizing initiatives that rely on data, in institutionalising the

behaviours that normalise the use of analytics. Conversely, employees view analytics as optional in case leaders are sceptical or indifferent leading to less organisational learning and minimal penetration of analytics. This difference underlines the way cultural changes start to take place with the managers believing in the utilization of analytics and not with technology itself.

A number of studies claim that leadership support is an intermediary that enhances the correlation between analytics ability and performance results. Chatterjee et al. (2022) demonstrate that the positive impact of big data capability and CRM capability is enhanced by leadership support in terms of the strategic sales performance. Their results suggest that the managers who proactively promote analytics develop a culture in which insights of analytics are not produced but implemented in strategy making. Relative to the organisations with poor managerial support, organisations with committed leaders have a high chance of transforming the capabilities of analysis to competitive advantage. This is in line with the argument in the general that leadership does not only influence adoption but also the efficacy of analytics to meet organisational objectives.

Although it is essential, managerial support is still limited in most cases by impediments associated with change management, resistance, and resources allocation. According to Smith (2025), the perceived complexity, the lack of skill in the workforce, and risk aversion are the fundamental factors that make the commercial banking leaders reluctant to embrace advanced analytics. On the same note, Kiu and Chan (2024) contend that managerial ignorance in understanding the usability of data analytics is a barrier to its implementation in the performance management system. Their results demonstrate how hierarchical schemes and lack of training establish the environment where managers see analytics as a burden instead of the factor that helps. These obstacles strengthen the fact that leadership resistance can be rather based not on contravening technology but on the doubt about its usage, its price, or its fit in the organisation.

Managerial backing is also vital in establishing trust in data as well as reducing uncertainty during decision making. Digital and analytical leaders can more readily read the insights with confidence and minimize ambiguity and stimulate data-driven decision-making in teams. Aftab et al. (2025) show that digital leadership improves the analytical power of big data and the willingness of the organisation to change under the influence of AI. They indicate that in cases where leaders are knowledgeable about analytics, they assist in reducing organisational

hesitation and enhancing the data credibility as a decision-making tool. Likewise, Wee et al. (2023) state that managerial confidence is developed with the help of small-scale analytical successes that are repeated and help build trust in the tools and the insights they generate.

2.6 Theoretical Framework

The theoretically grounded studies in the present research are "Information Processing Theory (IPT)" and the "Resource-Based View (RBV)" of the firm. Information Processing Theory texts that the performance of an organization is influenced by the quality, availability, and processing of information utilized in making a decision (Galbraith, 1973). In this perspective, the quality of data and analytics power improve the capability of an organization in minimizing uncertainty and enhancing strategic decisions. Quality data means there will be a correct interpretation, and the ability to analyse the data with the help of analytics will make the process quicker and more orderly. In addition, Resource-Based View is based on the premise that firm-specific resources such as analytics, data infrastructure, leadership assistance can lead to sustained competitive advantage in cases where they are both valuable and effectively utilized (Barney, 1991). The managerial support is thus placed in the perspective of a strategic initiative that facilitates analytics adoption and inculcates data-driven practices into the decision routines. In this way these theories warrant the investigation of the direct impacts of data quality, analytics capability, and managerial assistance on analytics adoption and effectiveness of strategic decision making.

2.7 Conceptual Framework

It can be stated that the conceptual framework is based on the assumption that organizational inputs that concern data and analytics have a direct impact on strategic decision outcomes. "Data Quality, Analytics Capability, and Managerial Support" are considered the independent variables in this study that indicate the main conditions in organizations that allow data-based decision-making. The dependent variables are "Business Analytics Adoption and Strategic Decision-Making Effectiveness" which indicate the adoption of analytics tools and their actual alignment to the strategic decision-making. Notably, adoption is not presented as an independent outcome variable which is affected by managerial and analytical conditions.

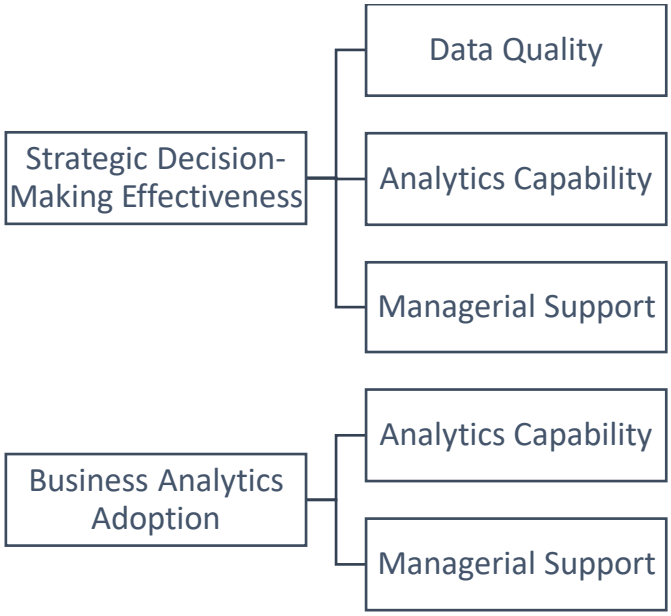


Figure 2: Conceptual framework

Independent Variables	Data Quality	The level of correctness, timeliness, totality and reliability of organizational data to be utilized strategically.
	Analytics Capability	The firms capacity to gather, process, examine, and reflect data through the use of analytical instruments and potentials.
	Managerial Support	How top and middle management inspire, invest in and propagate data-driven decision practices.
Dependent Variables	Business Analytics Adoption	The degree to which analytics are proactively applied in a process of making decisions in organizations.
	Strategic Decision-Making Effectiveness	The perceived quality, accuracy and timeliness of strategic decisions.

Figure 3: Definition of variables

The framework suggests that quality data enhances the effectiveness of decisions by increasing accuracy and reliability and confidence in the strategic decision-making. Good analytics power enables the organizations to manipulate information effectively enhancing timeliness and accuracy of decisions. It is anticipated that the managerial assistance will have a dual role in promoting analytics usage and enhancing decision-making behaviour by being leadership committed and assigning resources. Further the model enables the joint effect of data quality, analytics capability, and managerial support on the effectiveness of strategic decision making by having a multivariate relationship, and not via indirect pathways. This structure is conceptually clear and statistically sound. In this way following are the hypothesis that will be

tested.

H1: Data quality has a significant positive relationship with strategic decision-making effectiveness.

H2: Analytics capability has a significant positive relationship with strategic decision-making effectiveness.

H3: Managerial support has a significant positive effect on business analytics adoption.

H4: Analytics capability has a significant positive effect on business analytics adoption.

H5: Data quality, analytics capability, and managerial support collectively have a significant positive effect on strategic decision-making effectiveness.

2.8 Summary

The chapter has identified the key themes that lie behind data-driven strategic decision-making, explaining the particular role of data quality, analytics capability, and managerial support in the overall organisational results. The review contrasted divergent scholarly views and had a limitation of current research, which supports the need of integrated research. The chapter offers a consistent ground on the conceptual framework by combining theoretical knowledge and the results of the empirical research. These discussions present a strong rationale as to why there is a need to investigate combined effects of these variables on the effectiveness of strategic decision making.

Chapter 3: Research Methodology

3.1 Introduction

In this chapter, the decisions regarding the methodology that informed the design and implementation of the study are outlined. The chapter starts with laying down philosophical premises of the research and the theoretical premise on which the research methodology is based. It goes on to describe the design of the survey, sampling plan and the features of target participants. Specifications of the development of the survey instrument in terms of its construction, piloting and administration are also given. The chapter further explains the analytical steps that were employed in testing the hypotheses of the study.

3.2 POEMS

3.2.1 Research philosophy

The philosophy approach of this study is positivist, which is based on the desire to investigate the perceptions and behavioural reactions towards technology using quantifiable outcomes. Positivism presupposes that social phenomena can be studied with the help of the same principles in the natural sciences and that systematic observation and empirical testing are the main priorities in such a study. Saunders et al. (2009) have put forward positivism studies to be based on objective measurement and it does not seek subjective interpretation, a stance that is similarly reflected by Creswell and Creswell (2017), who observe that quantitative designs have advantages of clarity of constructs and reliability. In this position, the research considers variables like perceived usefulness, trust and adoption intention as measurable factors which can be statistically tested which allows more generalizable findings.

3.2.2 Ontology

The ontological standpoint that underlies this research is realism, especially the empirical one. The assumptions of realism presuppose that organisational and behavioural phenomena are not dependent on the researcher but can be measured systematically (Easterby-Smith et al., 2021). Based on this assumption, the investigation of the perceptions of AI-enabled platforms could be held as stable constructs that could be identified and compared among respondents. In comparison with relativist ontology, which puts an accent on the existence of multiple realities that are influenced by personal perspectives, realism holds that there is a fundamental existence, which can be reached through attentive measurements. This paper thus goes forward on the assumption that the attitudes towards AI, trust formation and adoption intention show a pattern which can be observed by the use of numerical data.

3.2.3 Epistemology

In line with this ontological perspective, the research follows an objectivist epistemology where knowledge is formed based on observable, measurable and externally verifiable data. According to Bryman (2016), objectivism focuses on maintaining a detachment between the researcher and the data to reduce bias, and also gives explicit guidelines on how to interpret the results. This position is opposed to constructivist epistemologies whereby emphasis is laid on the interpretations of the participants and negotiated meanings. Through objectivism, the study aims at producing results that come out of statistical associations and not interpretative considerations to ensure that the conclusions are based on the tendencies of the data.

3.2.4 Method

Considering the basis on positivism, the research takes the deductive approach to research. Deduction starts with the well-known theories, out of which a set of empirically testable hypotheses is developed and then analysed with the help of the empirical data (Saunders et al., 2009). This methodology is most appropriate in those studies which test relationships among predetermined constructs including the effect of trust or perceived usefulness in relation to technology adoption. It also has similarity with past technology-acceptance literature which has the theoretical grounding followed by empirical test (Venkatesh et al., 2003).

3.2.5 Strategy

The methodological approach adheres to a quantitative design with the aid of the structured online survey to gather data. This method is suitable to get perceptions of more people in a uniform format to give a statistical likelihood of comparison between the respondents. According to Pandey et al. (2023), structured surveys are efficient when one aims to investigate associations or measure hypotheses between pre-defined variables. In this study, the survey approach would be a viable way of measuring the attitudes, discerning the trends and establishing determinants of technology adoption among a group of students.

3.3 Research Strategy

3.3.1 Research design

The research design will be cross-sectional survey design, whereby data on participants will be collected at a given time. This type of design is best when the aim is to investigate the existing perceptions, behaviours or organisational practices and not to trace the changes over a long period of time. According to Saunders et al. (2009), cross-sectional surveys are effective to

enable the researcher to obtain standardised responses especially when they are investigating the relationship between more than two variables. In comparison to longitudinal designs, which presuppose repeated measurements and are more complex, the cross-sectional one allows gathering data timely but at the same time allows strong statistical analysis. The design is thus suitable when analysing the perception of professionals on the data quality, analytics capability and managerial support concerning the effectiveness of organisational decision-making.

3.3.2 Sampling Strategy

It employs the convenience sampling approach by targeting people employed in data driven or analytics oriented organisations. Probability sampling has greater generalizability, however convenience sampling is typically used in an organisational study when one must have a certain professional sample (Bryman, 2016). The sample will also consist of the members of the inclusion criteria, i.e. people engaged in decision-making, data analysis, management, or frequent analytics tools usage. The respondents will not be included in the study if they are not involved in organisational decisions or familiar with analytics practices because their answers would not offer any valuable insight into the constructs of interest. The expected number of participants is 100 to 150 participants, which is appropriate according to the principles of multivariate analysis and the ability to test the hypothesis (Pandey et al., 2023). This size is deemed to be adequate in determining patterns and creating statistically satisfying relationships between variables.

3.3.3 Participant Profile

The research is aimed at individuals working in the field of analytics in their day-to-day activities, such as managers, business analysts, data specialists, and operational employees who have to use analytical leads. These people are also in good position to consider how the quality of data and managerial situations influence the outcomes of decisions. The demographic factors will be used to identify important features of organisations, including job role, department, and years of experience, industry sector, and analytics exposure level. Their inclusion will allow identifying the trends within professional groups and assuming the findings in a more nuanced way, as Creswell and Creswell (2017) suggest. This design is necessary to make sure that the sample is representative of the wide spectrum of individuals who play a role in the production of the analytics-informed decision environments.

3.4 Data collection methods

3.4.1 Development of instrument

Through the survey instrument the researcher used existing items of measurement scales that are widely applied in organisational and information systems studies. The use of the past research helps to make sure that the constructs are based on the generally accepted definitions and increase the validity of the used methodology (Lamm et al., 2020). The variables, such as data quality, analytics capability, managerial support, analytics adoption, and strategic decision-making effectiveness, were operationalized with the help of multiple statements that encompassed important attributes of the variables as noted in the literature. To facilitate the consistency in the responses, the items were placed on a five-point Likert scale, between strongly disagree and strongly agree. This scale is also commonly suggested to be used in attitudinal studies due to the ease of the interpretation of its responses and the ability to make reliable statistical comparisons (Bryman, 2016). The design of the instrument is hence consistent with previous methodological directions and helps measure the relationships between the variables.

Independent Variables

- Data Quality
  - The level of correctness, timeliness, totality and reliability of organizational data to be utilized strategically.
- Analytics Capability
  - The firms capacity to gather, process, examine, and reflect data through the use of analytical instruments and potentials.
- Managerial Support
  - How top and middle management inspire, invest in and propagate data-driven decision practices.

Dependent Variables

- Business Analytics Adoption
  - The degree to which analytics are proactively applied in a process of making decisions in organizations.
- Strategic Decision-Making Effectiveness
  - The perceived quality, accuracy and timeliness of strategic decisions.

Figure 4: Definition of variables

3.4.2 Pilot Study

A pilot survey was carried out before the main survey was done and a small group of professionals with a background of working in data-driven settings took part. This was to test

the level of clarity of the wording, instructions and layout to the respondents. Pilot feedback forms part of determining items that may cause ambiguous or discrepancy (Saunders et al., 2009). With the comments received, some small adjustments were included e.g. simplification of phrasing, eliminating ambiguous words and changing the order of some of the items to enhance logical flow. Such modifications assisted in enhancing the reliability and face validity of the ultimate survey instrument.

3.4.3 Procedure of Data Collection

The online questionnaire was used to collect the data which was shared through the social media post and professional sites such as LinkedIn. This approach enables scholars to access participants who are located widely and this is common in organisational research (Creswell & Creswell, 2017). A short introduction to the purpose of the study was given to the respondents with the right to withdraw and guarantee of anonymity. The participation was purely based on voluntary basis and informed consent was taken prior to initiation of the questionnaire. By following steps the study served to guarantee that no ethical standards were violated during the data collection.

3.5 Data Analysis

The analysis of the data was done in an organized order that aimed at assessing the relationships hypothesized in the study. After the screening and cleaning of the survey data, the data was analysed in descriptive statistics to summarize the trends in the characteristics of respondents and give a rough idea about the variables (Bryman, 2016). This initial step assisted in detecting any anomalies, value gaps or outliers that should not be overlooked further during analysis. The internal consistency of the measurement scales was determined by computing the Cronbach alpha of each construct and it should be around 0.70 or above as shown in figure 4. Reliability testing is generally the most encouraged in quantitative research since it gives indications that the material included in the scale measures the same concept behind the scale (Lamm et al., 2020). The mathematical representation of cronbach alpha is below.

$$\alpha = \frac{k}{k-1k}(1 - \sum \frac{\sigma^2 i}{\sigma^2 T})$$

Model Testing and Decision Criteria	Reliability: Cronbach's alpha ≥ 0.70
	Correlation significance: $p < 0.05$
	Regression coefficients (β): tested using t-statistics
	Model significance: tested using F-statistic
	Explained variance: assessed using R^2 and Adjusted R^2
	Statistical significance threshold: $p < 0.05$

Figure 5: Testing criteria for the adopted models

Correlation analysis was performed following this to examine the strength and direction of association of the variables. Correlational methods provide initial information about potential associations without any premises that it is causal (Arkes, 2023). The essence of the analysis was the application of multiple regression that enables us to consider the predictive power of data quality, analytics capability and managerial support on analytics adoption and effectiveness of strategic decision-making as shown in figure 5. Regression is frequently applied in organisational research, as the model allows testing the joint and independent effect of a number of predictors in a single model (Creswell & Creswell, 2017). This methodology made analysis of the hypotheses of the study transparent and followed the recognised standards of the analysis.

Regression Model 1: Business Analytics Adoption

- $BAA = \beta 0+\beta 1MS+\beta 2AC+\varepsilon$
- Whereas
- BAA = Business Analytics Adoption
- MS = Managerial Support
- AC = Analytics Capability

Regression Model 2: Strategic Decision-Making Effectiveness

- $SDME=\beta 0+\beta 1DQ+\beta 2AC+\beta 3MS+\varepsilon$
- Whereas
  SDME = Strategic Decision-Making Effectiveness
- DQ = Data Quality
- MS = Managerial Support
  AC = Analytics Capability

Figure 6: Mathematical representation of regression models

3.6 Ethical Consideration

Ethical consideration has been adopted while conducting the research such as the participants were treated with fairness, respect and transparency. The research adhered to accepted ethical

standards, giving a primary focus to voluntary participation and offering the respondents with an adequate information regarding the purpose of the study and the use of their data (Saunders et al., 2009). Prior to starting the survey, a statement about informed consent was introduced which explained the lack of risks, voluntary withdrawing and anonymous response processing. No personal identifiers were gathered and this minimized the chances of identifying any response with an individual. This was in accordance with the suggestions that anonymity assists in safeguarding the participants against unintended harm (Bryman, 2016). All data were kept in a safe location in digital files with passwords, and only the researcher and the academic supervisor had access to them in compliance with the data protection requirements and GDPR principles. Ethical implication of honesty and transparency was also taken into account in the study. There were no deceptive practices involved and the respondents were on the notification of the nature of the study and the nature of questions that will be asked (Creswell & Creswell, 2017). The research aimed to uphold integrity, protect the participants, and adhere to every policy in the institutional ethics by meeting these standards.

3.7 Summary

The chapter has provided information on the manner in which the study was methodologically organized. It also described the philosophical inclination upon which the study was being conducted and the role of a positivist and deductive position on the rationale behind the application of quantitative approaches. The chapter also explained the cross-sectional survey design as to why it was the best to use to collect similar information about individuals engaged in analytics-related organisational functions. To show reliability and clarity, the process of the development and refinement of the survey tool and the pilot testing process were discussed. The process of data collection and the ethical considerations were also defined.

Chapter 4: Findings and Results

4.1 Introduction

This chapter is the reporting of the empirical study results on the basis of quantitative analysis of the data obtained (111 respondents). It presents the descriptive statistics, reliability tests and inferential tests of the relationship of data quality to analytics capability, managerial support, business analytics adoption, and the success of strategic decision-making. The findings are presented in a clear and objective way straight to the point as the research objectives and hypotheses.

4.2 Descriptive Statistics

The sample size (N = 111) is the demographic representation of participants who are active within the organisational setting where the use of analytics becomes more and more topical. There was a gender distribution where 82 respondents (73.9) were males, 28 (25.2) females, and one respondent did not want to disclose as shown in figure 7. The gender value (M = 1.27, SD = 0.47) in table 1 shows that there is not a lot of variance, which implies the presence of a rather focused demographic organization. Though male respondents are dominant in the sample, this behaviour is consistent with the previous findings in the analytics-intensive and technology-demanding organisational positions (Saunders et al., 2009).

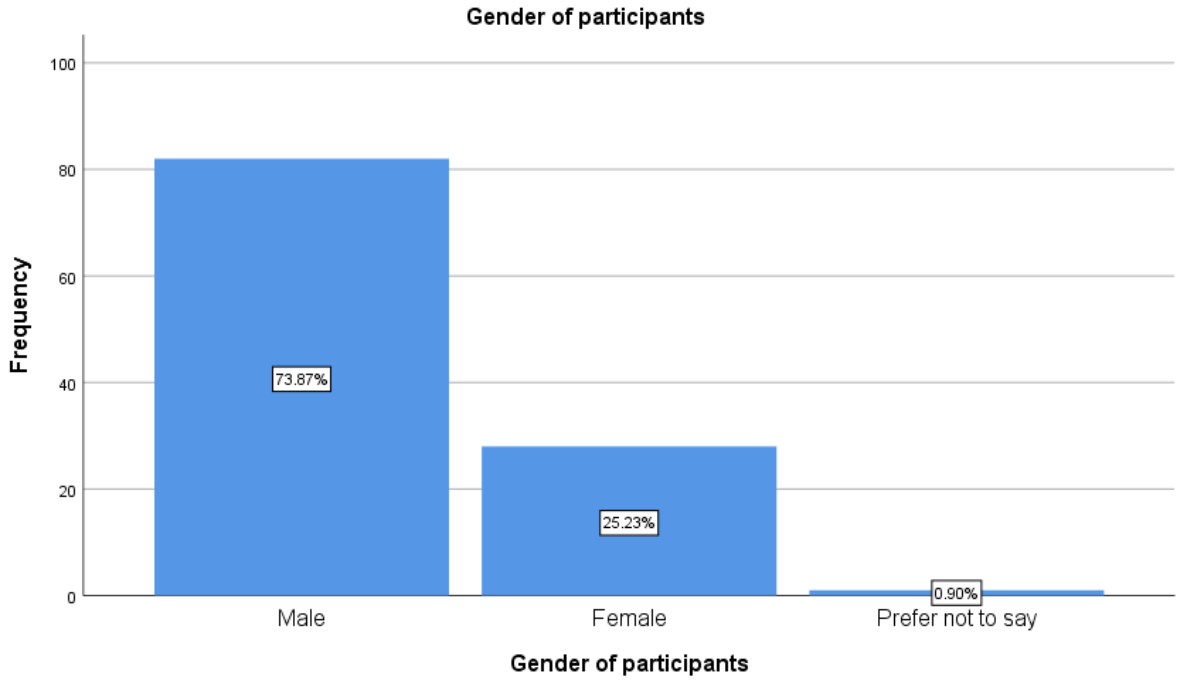


Figure 7:"Gender of participants"

The age structure also portrays a fairly young respondent base. The respondents between the

ages of 18-24 years (51.4% (n = 57)) and those between the ages of 25-34 years (38.7% (n = 43)) comprised 51.4 and 38.7 percent of all respondents respectively as shown in figure 8. A total of 90.1% of the respondents were less than 35 years old. This skew towards early-career professionals is confirmed by the mean age category score (M = 1.59, SD = 0.69) as shown in table 1. The age composition is applicable, and younger employees are more likely to be more open to analytics tools and digital decision-making practices, which may contribute to the positive attitudes toward analytics implementation (Hair et al., 2018).

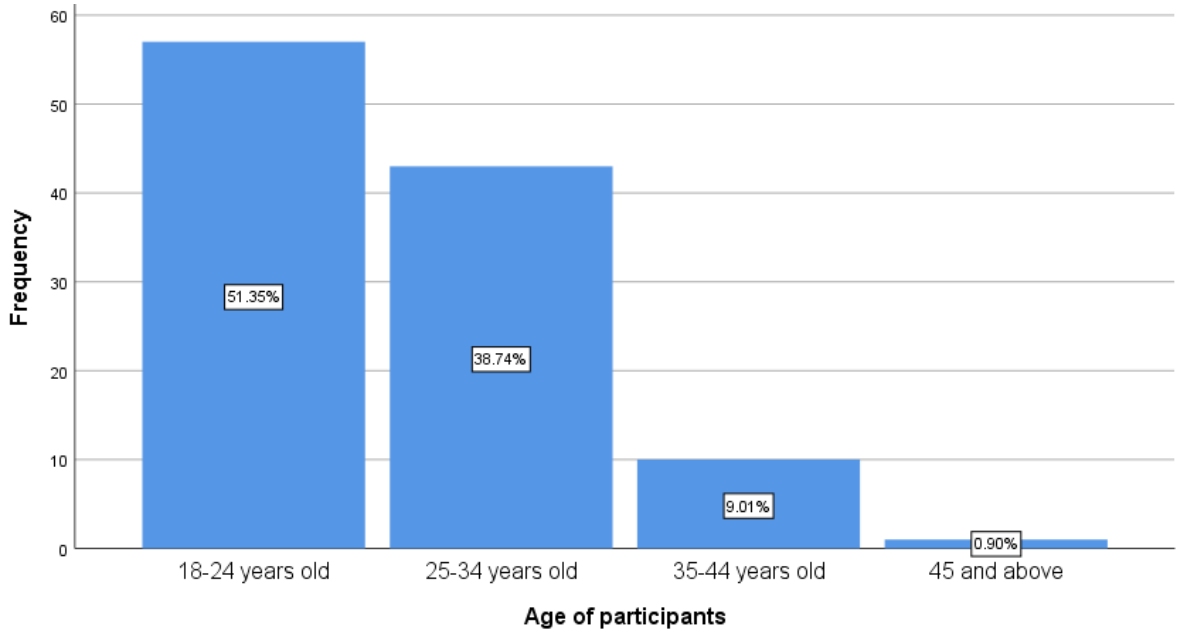


Figure 8: "Age of respondents"

The respondents had a high educational level. Most of them had a master degree (56.8, n = 63), then there were those with bachelor degree (31.5, n = 35). Doctoral qualifications were found to be 5.4% (n=6) and professional certifications were found to be 6.3% (n=7) (Figure, 9). The average education level (M = 1.86, SD = 0.78) shows that it is focused on postgraduate levels. The profile of education corroborates the analytical richness of the replies, since the level of higher education has a positive correlation with analytical thinking and the ability to make decisions supported by evidence (Bryman, 2016).

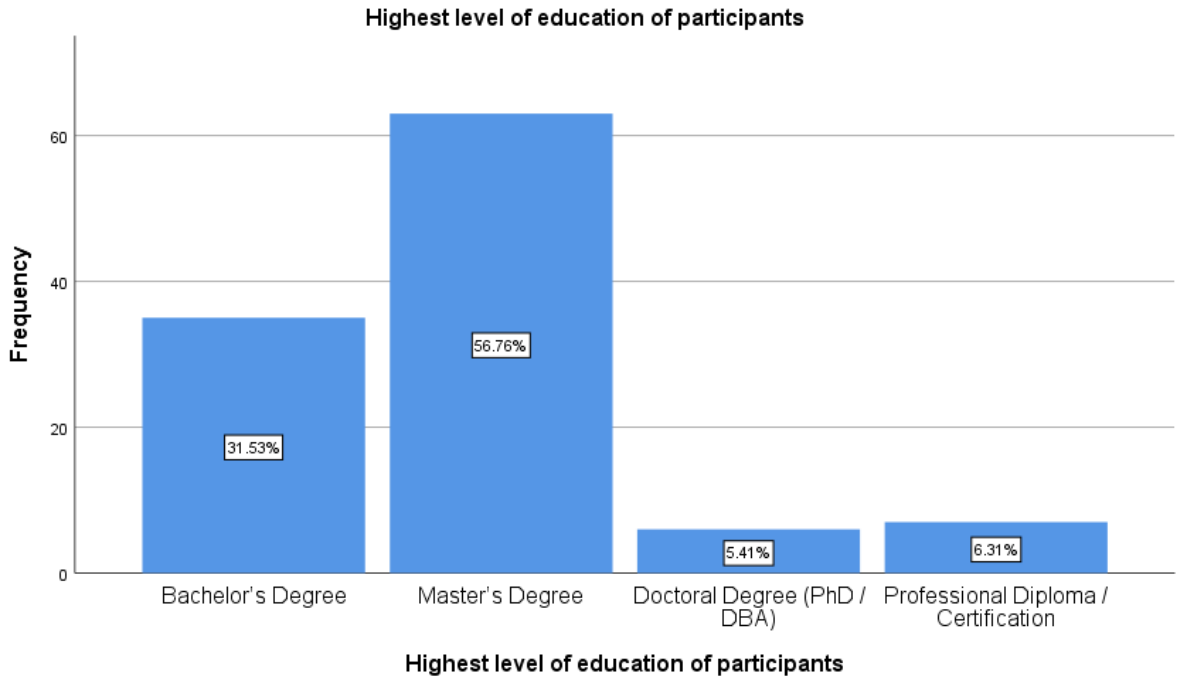


Figure 9: "Education of participants"

The representation of the industries and the departments also show significant diversity. Most of the respondents were attracted in the retail/wholesale (31.5%), information technology/software (30.6%), and manufacturing industries (13.5%). IT and data analytics had 31.5% (n = 35), marketing/sales 28.8% (n = 32), and operations/supply chain 12.6% (n = 14) in the departments (Figure, 10). There is a mean industry score (M = 3.24, SD = 1.68) and department score (M = 3.60, SD = 1.76), which indicates the existence of a wide dispersion and increases the generalizability of analytics-related results in a functional context (Someh et al., 2023).

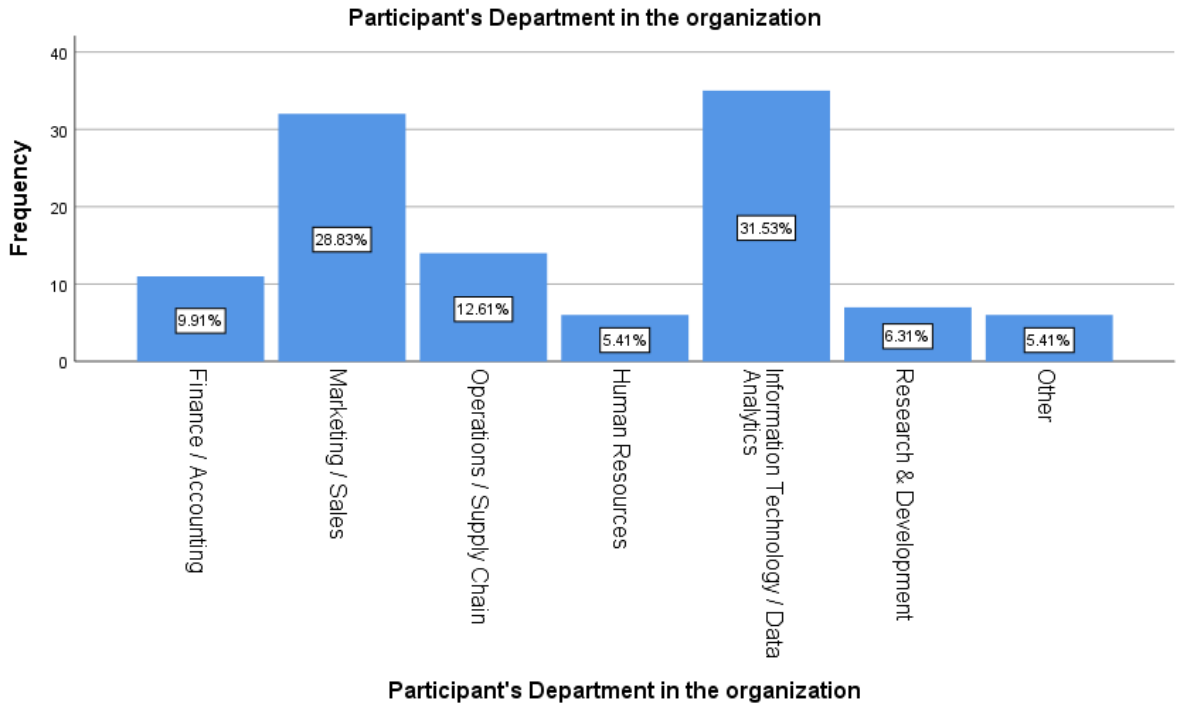


Figure 10: Participants department of the organisation

The key research variables are evaluated positively at all times in table 1. The mean scores in the data quality (M = 3.83, SD = 0.96), analytics capability (M = 3.81, SD = 1.00), managerial support (M = 3.83, SD = 1.00), business analytics adoption (M = 3.88, SD = 1.04), and strategic decision-making capability (M = 3.93, SD = 1.03) show moderate-to-high agreement levels among the respondents. Significantly, the constructs show excellent internal reliability, with Cronbachs alpha values between 0.942 and 0.965, being higher than the advised levels in social science studies (Hair et al., 2018). These findings justify the strength of the measurement scales and offer a sound foundation of further inferential analysis.

Table 1: Descriptive statistics

Descriptive Statistics						
	N	Minimum	Maximum	Mean	Std. Deviation	Cronbach alpha
Gender of participants	111	1.00	3.00	1.2703	.46605	
Age of participants	111	1.00	4.00	1.5946	.69254	
Highest level of education of participants	111	1.00	4.00	1.8649	.78029	
Participant's current position in the organization	111	1.00	5.00	2.8559	1.11882	
Participant's industry	111	1.00	8.00	3.2432	1.67991	
Participant's Department in the organization	111	1.00	7.00	3.6036	1.75954	
Participant's total professional experience	111	1.00	5.00	2.2613	.93139	
Participant's rating of their organization's overall use of business analytics	111	1.00	5.00	3.4234	1.01399	
Participant's familiarity with analytics tools	111	1.00	5.00	2.9640	1.30682	
Data Quality	111	1.00	5.00	3.8324	.95633	0.942
Analytics Capability	111	1.00	5.00	3.8108	1.00357	0.952
Managerial Support	111	1.00	5.00	3.8342	1.00104	0.958
Business Analytics Adoption	111	1.00	5.00	3.8811	1.04076	0.965
Strategic Decision Making Effectiveness	111	1.00	5.00	3.9333	1.03092	0.954
Valid N (listwise)	111					

4.3 The relationship between data quality and strategic decision-making effectiveness in data-driven organizations

The connection between the quality of the data and the effectiveness of strategic decisions was investigated to answer the first research question and Hypothesis H1. The findings indicate that there is a strong and positive correlation between the two variables and the Pearson correlation coefficient of r = 0.828 in table 2 presents a very high level of linear association, which implies that the improvement of the quality of data is closely related to the improvement of the efficiency of strategic decision-making. The significant level of the relationship (the significance level p = 0.000) proves that this is a statistically significant relationship at the level of 1 percent and gives a strong empirical evidence to H1. Practically, this means that the scale of the correlation is that data quality is not a peripheral factor, but the primary focus of strategic performance in a data-driven organization (Hair et al., 2019).

Table 2: Correlation analysis between data quality and strategic decision making effectiveness

		Strategic Decision-Making Effectiveness
Data Quality	Pearson Correlation	.828**
	Sig. (2-tailed)	.000
	N	111
**. Correlation is significant at the 0.01 level (2-tailed).		

The Quality of data in term of accuracy, completeness, timely and consistency seems to enhance the skill of decision-makers to analyse the complex environment and consider strategic

options with more confidence. The inherent nature of the observed correlation is that as the decision-makers have placed their faith in the underlying data, the probability that they will place their trust in the output of the analytical process instead of the intuition as a single force is likely to happen. This observation is consistent with the previous literature that suggested low data quality creates a noise and ambiguity in decision making, thus undermining strategic integrity and predisposing decision making to make suboptimal decisions (Redman, 2016). Conversely, trustworthy data will decrease information asymmetry and increase the transparency of the strategic signals that can allow more decisive and proactive decision-making.

The finding further goes hand in hand with empirical investigations that locate data quality as a pillar of analytics-based decision-making. As an example, research in analytics and information systems literature will invariably show that excellent analytic tools cannot be used to create value in situations where the data underpinning them is faulty (Wixom et al., 2013). The fact that correlation was very high in this study is an indication that data quality might have a more direct impact on decision effectiveness than other enhancing capabilities like analytics infrastructure or management support espoused, especially in companies with already existing data-driven practices. This supports the point that data quality is not a technical issue but a strategic one to be handled (Davenport & Harris, 2017).

4.4 Assessing how organizational analytics capability influences the accuracy and timeliness of strategic decisions

The second research question was whether the organizational analytics capability has an effect on the accuracy and promptness of strategic decision-making as proposed in the Hypothesis H2. According to the empirical findings, the relationship between analytics capability and strategic decision-making effectiveness is very strong and the Pearson correlation coefficient is at $r = 0.854$ with a sample of 111 respondents. This is much higher than the traditional criterion of strong association and this value indicates analytics capability as an important organizational driver of successful strategic decisions. The correlation is significant (1 percent, two-tailed, $p = 0.000$), which means that the given correlation occurs with high probabilities and gives a clear evidence in favour of H2.

		Strategic Decision-Making Effectiveness
Analytics Capability	Pearson Correlation	.854**
	Sig. (2-tailed)	.000
	N	111
**. Correlation is significant at the 0.01 level (2-tailed).		

The correlation magnitude indicates that organizations that have properly developed analytics are in a better position to come up with time and correct strategic insights. The analytics capability is generally an indicator of the synergies between the analytical tools, human resource, data integration processes and conversion of analytical outputs into managerial action. Having these factors aligned enables decision-makers to cease doing descriptive reporting to predictive and prescriptive reporting to enable quicker responses to environmental changes. A previous study also highlights the importance of analytics capability to increase decision speed and decrease the use of fragmented information and manual interpretation, and, at the same time, the accuracy of the decision made due to a systematic and evidence-based assessment.

The size of the correlation found in the present research, in comparison to previous empirical studies, supports conclusions made by prior research empirical studies which identified analytics capability as a more effective predictor of the quality of decisions than the investment in technology. The organizations with superior analytical tools, yet they do not have analytical skills or interpretive routines tend to fail to transform data into actionable strategy (Wamba et al., 2017). Contrarily, companies that exhibit developed analytics are better at enabling real-time data into strategic deliberations, which eliminates delays and decreases cognitive bias in the executive decision-making process. The large value of the correlation coefficient ($r = 0.854$) shows that analytics capability contributes to reducing the gap between the availability of information and strategic action to a considerable extent. Strategically, the study has confirmed that analytics capability is a dynamic capability that improves organizational responsiveness. Analytics capability allows to make accurate strategic decisions as well as schedule them, which is critical in uncertain and volatile markets conditions, enabling to analyse the scenario rapidly and predict performance (Teece, 2018).

4.5 Determining the effect of managerial support for data-driven practices on the adoption and effectiveness of business analytics in strategic decisions

The third research objective was to test the effect of managerial support of data-driven practices and analytics capability in the adoption of business analytics in strategic decision-making.

Regression findings indicate that the model has a high explanatory power with the overall correlation coefficient of R = 0.942 and R 2 = 0.887. It means that managerial support and analytics capability explain 88.7 percent of the variance in the adoption of business analytics. The good value of the adjusted R2 at 0.885 also underscores the strength of the model, which implies that there is not much inflation based on the sample size, and it reaffirms the applicability of the elements of leadership and capability in adopting analytics (Hair et al., 2019). Additional model evidence is obtained by the results of ANOVA. The regression model gave F-statistic of 424.483 and the significance value of p = 0.000 and this indicates that the joint effect of managerial support and analytics capability on business analytics adoption is statistically significant. The fact that the residual mean square is relatively low (0.125) implies that the predictors do a meaningful job of explaining the relevant organisational drivers of analytics adoption. The results are consistent with the existing literature on the importance of top management commitment and analytical readiness in the implementation of analytics within the strategy (Davenport & Harris, 2017).

Table 3: Regression analysis for business analytics adoption

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.942a	.887	.885	.35286
a. Predictors: (Constant), Managerial Support, Analytics Capability				

ANOVAa						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	105.703	2	52.852	424.483	.000b
	Residual	13.447	108	.125		
	Total	119.150	110			
a. Dependent Variable: Business Analytics Adoption						
b. Predictors: (Constant), Managerial Support, Analytics Capability						

Coefficientsa						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.062	.135		.456	.049
	Analytics Capability	.511	.083	.493	6.124	.000
	Managerial Support	.488	.084	.470	5.836	.000
a. Dependent Variable: Business Analytics Adoption						

A review of the regression coefficients shows that there is a strong support of the hypotheses

H3 and H4. The analytics capability has a positive and significant impact on the business analytics adoption with a standardized beta coefficient of β = 0.493, t value 6.124 and p =0.000. This implies that organisations that have robust analytical structure, have talented staff and routines of interpretation have higher chances of adopting analytics. In the same manner, the positive impact of managerial support is strong and the results are β = 0.470, t = 5.836, p = 0.000, which implies management support in form of encouragement, resource allocation and advocacy is a key factor in legitimising analytics-based practices in organisations. By comparison, the more pronounced beta of analytics capability implies that there is a minor difference between technical and human preparedness and managerial power in influencing adoption. The close value of the two coefficients, however, is an indication of their complementary character. Existing studies claim that despite the developed capabilities in analytics, they can be poorly utilized unless they are supported by well-intentioned management, whereas weak analytical grounds can be addressed by strong leadership only (Wamba et al., 2017). The results thus indicate reinforcing relationship, in which management support enhances the successful application of analytics ability, resulting in an increase in the adoption levels.

4.6 Examining the collective impact of data quality, analytics capability, and managerial support on overall strategic decision-making performance

The fourth research objective explored how the combination of data quality, analytics capability and managerial support affects the effectiveness of strategic decision-making. The overall relationship between the regression model shows a high degree of association between the joint predictors and decision-making performance because of the high multiple correlation coefficient of R = 0.874. The R2 value 0.764 implies that the combination of the three variables explains the 76.4% of the variance in decision-making effectiveness in strategic decision making and the adjusted R2 value 0.757 indicates the consistency of the model after the sample size and the number of predictors. This is believed to be a significant degree of explanatory power in an organization research, which supports the claim that strategic decisions are influenced by an integrated analytics environment and not by individual considerations (Hair et al., 2019).

The statistical significance of the model is also confirmed by the results of the ANOVA. The regression generated the F-statistic of 115.364, and the level of significant p = 0.000 with a regression model, which proved that the joint predictors significantly enhance the prediction of the effectiveness of strategic decision-making when compared to a null model. The relatively

low residual mean square (0.258) implies that there is not much unexplained variance, and the data, analytics, and managerial support, in turn, are the combination of the main drivers of analytics-based strategic decision-making. This result falls into line with systems-oriented thinking, which holds that the effectiveness of a decision is developed by the interplay between data, tools, and leadership, and not by a single resource (Davenport & Harris, 2017).

Table 4: Regression analysis for strategic decision making effectiveness

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.874a	.764	.757	.50796
a. Predictors: (Constant), Managerial Support, Data Quality, Analytics Capability				

ANOVAa						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	89.299	3	29.766	115.364	.000b
	Residual	27.608	107	.258		
	Total	116.907	110			
a. Dependent Variable: Strategic Decision Making Effectiveness						
b. Predictors: (Constant), Managerial Support, Data Quality, Analytics Capability						

Coefficientsa						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.293	.205		1.428	.006
	Data Quality	.325	.118	.302	2.750	.007
	Analytics Capability	.293	.156	.285	1.878	.000
	Managerial Support	.333	.120	.324	2.768	.007
a. Dependent Variable: Strategic Decision Making Effectiveness						

A review of the standardized regression coefficients reveals that all the three predictors have statistically significant positive effects, therefore, confirming H5. The most standardized effect is managerial support ($\beta = 0.324$, $t = 2.768$, $p = 0.007$) and then there is data quality ($\beta = 0.302$, $t = 2.750$, $p = 0.007$). There is also a positive contribution of analytics capability ($\beta = 0.285$), which shows that it plays a significant role in improving the accuracy and timeliness of the decision made, though this impact is not that strong when the leadership and the quality of the data are taken into consideration. These results imply that tools and abilities of analytics are necessary; however their strategic merit is optimised when managers understand the importance of these methods and when the data of the data proved to be valid and pertinent. Comparatively, the findings are reminiscent of the literature that underscored complementarities of analytics-related resources. Data of high quality that lacks managerial confidence could not affect the decisions, and a robust leadership supporting it cannot

substitute poor data underpinning (Wamba et al., 2017; Teece, 2018).

4.7 Summary

This chapter has examined the data in terms of descriptive and inferential statistical methods in order to test the hypotheses put forward. The results showed that there were strong, positive, and statistically significant correlations between the major constructs and data quality, analytics capability, and managerial support are relevant to induce high effectiveness in strategic decision-making. The combined effect of these variables on the adoption of business analytics and decision outcomes is further exhibited by the results of the regression. The findings give solid empirical evidence to the topic conceptual framework.

Chapter 5: Discussion on the Findings and the Results

5.1 Introduction

In this chapter, the empirical results are interpreted concerning research objectives, hypotheses, and theoretical basis of the study. The discussion will be based on the "Information Processing Theory (IPT) and the Resource-Based View (RBV)" to determine whether the expected results have been achieved, the manner in which the findings compare with the existing literature and what the research will teach us about making strategic decisions using data.

5.2 Data Quality and Strategic Decision-Making Effectiveness (Objective 1)

The initial research question tested the connection between quality of data and effectiveness of strategic decision-making. Using the IPT, the research hypothesized that quality of the data would lead to lesser uncertainty, greater accuracy in interpretation, and better confidence of the managers in their decisions. This expectation was solidly supported by the findings as a strong positive correlation between the quality of data and effectiveness of the decision was observed.

This fact aligns with the previous research, which places data quality as a core facilitator of decisions made with analytics (Khong et al., 2023; Hurbean et al., 2024). High accuracy levels, complete and timely levels seem to enhance the results of the analysis besides the desire of managers to use the results in strategic decisions. This research confirms the importance of data quality as a strategic issue, unlike other studies that consider data quality as a technical issue, which proves that even sophisticated analytics systems would fail without high-quality data. In the RBV perspective, the quality of data is a resource that is valuable and hard to imitate. Companies that invest in data quality control and alignment are also better placed to produce quality insights hence yielding quality decisions. This goes to strengthen the argument that data quality is not only an operational input, but a strategic asset, which is a corner stone in competitive decision-making.

5.3 Analytics Capability and Accuracy and Timeliness of Decisions (Objective 2)

The second objective involved measurement of organisational analytics capability impact on effectiveness of strategic decision-making. Analytics capability conceptualisation was based on RBV and was taken to be a combination of technical infrastructure, skilled personnel and analytical processes. The research anticipated a positive impact relationship with a high degree of positivity between the analytics and how it is able to convert raw data into actionable information. This expectation turned out to be true, with the findings revealing a very high level of association between analytics capability and effectiveness of strategic decision-

making. This is in line with the existing empirical literature that has shown that analytics capabilities have a positive effect on forecasting accuracy, responsiveness, and strategic alignment (Mishra et al., 2019; Su et al., 2022). Nevertheless, this research paper builds upon the current body of literature by empirically establishing the fact that the analytics capability directly enhances the effectiveness of the decisions, but is not working only through the intermediate results, like agility or innovation.

On the IPT perspective, the capacity of analytics enhances the information processing capacity of an organisation, which would be able to handle complex and dynamic environments. Companies of high analytics ability are in a better position to work on massive amounts of data, identify patterns, and result in timely information, which has a direct effect on the accuracy of strategic decisions. This helps to argue that analytics capability is not a matter of choice, but rather a key to successful strategic management in data intensive environment.

5.4 Managerial Support, Analytics Capability, and Business Analytics Adoption (Objective 3)

The third objective studied the impact of managerial support on the adoption of business analytics, as well as the influence of analytics capability. Managerial support was anticipated to be an enabling critical element in the study because it forms organisational attitudes, resource allocation, and data-driven practices legitimisation. The results of the regression proved that managerial support as well as analytics capability have strong positive impacts on analytics adoption. The result is consistent with the literature on leadership and its significance in analytics projects, which has stressed the significance of executive sponsorship (Chatterjee et al., 2022; Smith, 2025). The absence of managerial support may place analytics usage in restraint even in the presence of technical capabilities, which once again supports the notion that cultural and behavioural aspects are of as much significance as technological ones. It is interesting to note that the findings indicate that the analytics capability and the managerial support are not independent of each other.

The results expand the available literature by providing empirical evidence that the support provided by the managers' does not just moderate the outcomes of analytics but it is directly associated with the adoption of analytics. This brings up an essential practical point analytics infrastructure will not be who or what it could be without leaders actively encouraging and believing in data-driven methods. This addition makes the behavioural aspect of RBV stronger, presenting the way in which leadership strengths complement technical resources.

5.5 Collective Impact on Strategic Decision-Making Effectiveness (Objective 4)

The last objective examined the interaction between the quality of data, analytics ability, and managerial support and the effectiveness of strategic decision-making. This study theorised the existence of a strong positive impact of these variables, as the nature of decision environments based on analytics requires interdependence. This hypothesis was highly supported by the outcomes, and the combined model accounted a significant percentage of variance on the effectiveness of decision-making. Notably, no one of the variables overtook the model, which implies that balance is the foundation of the performance of strategic decisions, and that no individual factor is over-relied on. This observation is consistent with systems-based concepts of analytics, which believes that value is generated by complementarities of data, technology, and leadership (Teece, 2018). Theoretically, this compound effect helps to strengthen IPT and RBV. IPT describes the combination of these factors to increase the capacity of processing information, whereas RBV describes the combination of these factors to form a resource bundle that is valuable and rare and also hard to copy. This is one of the theoretical contributions of the work.

5.6 Summary

The study empirically confirms an integrated model, which concurrently investigates the aspects of data quality, analytics capability, and managerial support, which is a void in the existing literature, which tends to discuss the variables separately. Second, it enhances theoretical integration because of integrating both IPT and RBV to explain the effectiveness of strategic decision-making in data-driven organisations. Third, the research mentions the strategic, rather than technical, significance of data quality and leadership support. The results provide a practical understanding, as they showed that the success of analytics does not rely much on the implementation of advanced tools but aligning the integrity of data and organisational ability, as well as managerial dedication. This is a holistic view that can give a more realistic insight into the success of data-driven strategies in practice.

Chapter 6: Conclusion and Recommendation

6.1 Introduction

It connects the new results with the previous literature, analyses the success of the dissertation in achieving its goals, points out the weaknesses, and provides specific suggestions to practice and further research.

6.2 Conclusion

This dissertation aimed to explain the relationship between data quality, analytics capabilities and management support and how they are important in supporting strategic decision-making application in data-driven organisations. Empirically, the research provided powerful and consistent results that both data quality and analytics capability were correlated with a very high value ($r = 0.828$ and $r = 0.854$), and that the three predictors combined, to describe a large portion of the fraction of variance in decision effectiveness ($R2 = 0.764$). The results achieved meet the aim of the research and support the main hypotheses (H1-H5) that the conceptual model based on the "Information Processing Theory (IPT)" and the "Resource-Based View (RBV)" can help to gain insight into the analytics-led strategy (Galbraith, 1973; Barney, 1991; Khong et al., 2023; Mishra et al., 2019).

The empirical trend gives a strong theoretical fit. According to IPT, organisations expand their information processing capacity to minimize uncertainty; in this case, analytics capacity served precisely as a capacity to process data in time and decision relevant signals (Galbraith, 1973; Teece, 2018). RBV augments this by placing the quality data, analytics capabilities and managerial dedication as a prized, rare and complementary resources whose combination result in enduring decision benefits (Barney, 1991; Davenport & Harris, 2017). The current findings are both extensions of the two theories, as they demonstrate that the alignment of these resources - not any one investment - brings the most significant returns on the effectiveness of strategic decision-making, which is consistent with the recent empirical studies on resource complementarities in analytics (Su et al., 2022; Hurbean et al., 2024).

The peculiar value of the study is the quantification of those complementarities empirically in a single model. Previously, the quality of data, capability or leadership, and data quality were typically considered separately; the dissertation attempts to test the two together and show that managerial support and data quality have similar standardized impacts on the analytics capability as standardized impacts in the entire model (Ghasemaghaei & Calic, 2019;

Chatterjee et al., 2022). In practice, the wording cannot be misunderstood: the decision to invest in algorithms or platforms without investing in data integrity and leadership alignment will harm the under-realisation of strategic value (Ilmudeen, 2021; Wamba et al., 2017).

6.3 Recommendations

The study provide practical recommendations. To begin with, organisations ought to consider data quality as a strategic asset: they need to institute ongoing data management, invest in cleaning and lineage solutions, and implement metrics to ensure that data quality is visible to business leaders. Second, develop an analytical capability as a whole- include investments in cloud and tooling with specific hiring and up-skilling, and develop analytics operating models (centres of excellence, embedded data scientists) that connect insights to strategy. Third, ensure visible managerial sponsorship: leaders are to become evidence-based decision rituals and apportion analytics budget to analytics roadmaps, and incentivise analytics usage with appraisal systems, thus reducing resistance and normalising the use of data.

To educators and policymakers, the research suggests that data literacy and leadership should be increased in the curriculum. The industry data standards and privacy frameworks, which minimize the friction in cross-firm analytics and protect citizens, can be backed by the public policy. The executive programmes and higher education need to focus on interpreting outputs and change management to ensure leaders are able to make sense of algorithmic output and guide adoption.

6.4 Limitations

Realistic constraints and restrictions are also presented in the study. The study has a cross-sectional survey format, which only records perceptions and relationships at a particular point in time and therefore has limitations to the assigning of causality and long-term effects, longitudinal design or quasi-experimental would be more effective to reveal the temporal cause and effects of the study, as well as the learning processes (Saunders et al., 2009). The purposive sample (N = 111) was biased towards younger people and men and focused on specific sectors (retail, IT), which can weaken the applicability of findings to older generations of leadership or on areas with stringent regulation (banking, state administration) (Smith, 2025). Self-reported measures open the chances of common method bias although it has high Cronbach-alpha reliability; archival performance measures and objective adoption measures should be used to triangulate the results in future studies (Podsakoff et al., 2003).

6.5 Future directions

The nature of capability building and managerial learning would be tested in longitudinal research; mixed-method research would be used to investigate how certain governance practices would influence diagnosis and trust in analytics (Ghasemaghaei & Calic, 2019). Relationships between different sectors and countries would provide boundary conditions, i.e., are different industries highly regulated and have different adoption patterns (Khan et al., 2024). Further, cause claims would be significantly fortified by experimental or field intervention studies to determine performance change following governance, capability and leadership interventions.

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Appendix 1: Reflection

Introduction

This reflection essay is a critical analysis of my learning experience during the dissertation, and the broader academic programme. Applying the "Reflective Cycle devised by Gibbs" as a prospective guide, it examines my experiences, difficulties and self-growth in the course of the research. The reflection is based on the understanding of how theoretical knowledge, methodological skills and critical thinking developed during the writing of the dissertation. It also brings out my experience in the programme that has led to my academic confidence and professional competence preparedness.

Description

This dissertation is the result of my academic experience in the programme and is indicative of how I have been involved with the data-driven decision-making process, business analytics, and strategic management. The paper examined the impact of data quality, analytics capability, and managerial support on effectiveness of strategic decision-making in contemporary organisations. During the process, I performed topic refinement, literature synthesis, quantitative data collection, statistical analysis as well as academic writing. The theoretical framework and the instrumentation required to plan and carry out this research was offered by the programme, and the dissertation itself was a practical examination of the acquired knowledge, skills and discipline during the period of my studies.

Feelings

I was motivated and overwhelmed at the beginning. The increased applicability of business analytics to strategic decision-making was highly congruent with my academic interests, but I was initially scared by the scope of the dissertation. I was specifically worried about the ability to create a logical idea base and perform statistical analysis on my own. Later on in the course of the research, these sentiments changed to confidence and ownership. The accomplishments created in completion of data analysis and interpretation brought about achievement especially where the findings made sense based on theory. Towards the end, I was more confident in my analytical skills and more inclined to think critically with academic literature, which was a huge personal and academic change.

Evaluation

There were a number of factors that went well with the dissertation process. The incorporation of the Information Processing Theory, as well as the Resource-Based View, presented a powerful theoretical foundation, which enabled the study to shift out of the superficial analysis.

The quantitative design facilitated easy testing of hypothesis and the high reliability scores as indicated by the Cronbach alpha values more than 0.94 values supported the reliability of the measurement tools. Nevertheless, it was not without difficulties. The management of time was now and then a challenge especially balancing between data analysis and writing deadlines. Also, the use of survey data that was self-reported restricted the organisational insight. Nevertheless, these limitations did not prevent the presence of the organized assistance of the programme including, in particular, the research methods and modules connected with analytics, which reduced most of the challenges.

Analysis

Analytically, reflectively, this dissertation enhanced my learning on the translation of theory into practice. The Information Processing Theory made me understand the reason why organisations invest in analytics in order to cope with uncertainty and the Resource-Based View enabled me to understand the importance of data quality, analytical competency, and leadership support as strategic resources and not technical ones. The programme strengthened the use of methodological rigour, ethical research practices, and the critical analysis of the findings. Another thing that I have learnt is that empirical findings are seldom found in isolation; they have to be viewed in a conversation with the earlier research, inconsistencies, and situational limitations. This critical thinking, evidence-based reasoning, and applied research focus on the programme provided the result of this analytical maturity.

Conclusion

Reflectively, the dissertation was not a just a scholarly necessity but a growth process that transformed my perspective in solving complicated situations. I have now noticed that, successful research involves endurance, flexibility, and reflexivity. The programme has provided me with transferable skills, such as data analysis, academic writing, structuring and critical evaluation, that are not only applicable within the academic sector but also within the workplace. Although the research met its goals and made a small contribution to the current body of knowledge, the study also demonstrated the aspects of my development that can be further enhanced, especially the level of advanced analytics method and mixed-method research design.

Action Plan

In the future, I will expand the competencies that I have gained in this programme. In my professional field, I will enhance my analytic capabilities by working with more sophisticated tools like python based analytics and dashboard visualisation systems. In academics, my goal

is to stay active in the existing research of data-driven strategy and decision science, possibly by continuing my education or conducting applied research. In case of such a research in the future, I would embrace a longitudinal or mixed method design to measure more organisational dynamics. However, most importantly, I will still be practicing reflectively as advised by Gibbs cycle in order to be guaranteed of unending learning and improvement in both academic and career activities.

Appendix 2: Questionnaire

Demographic Information

Kindly tick the suitable option below

Gender	Male		Female	
	Prefer not to say		Other	

Age

18-24 years old	
25-34 years old	
35-44 years old	
45 and above	

What is your highest level of education completed?	
Bachelor's Degree	
Master's Degree	
Doctoral Degree (PhD / DBA)	
Professional Diploma / Certification	
Others	

Which of the following best describes your current position in the organization?	
Senior Management (e.g., CEO, Director, Head of Department)	
Middle Management (e.g., Manager, Team Lead)	
Operational / Functional Staff	
Data / Analytics Specialist	
Other	

Which industry best represents your organization?	
Manufacturing	1
Retail / Wholesale	2
Banking / Financial Services	3
Information Technology / Software	4
Healthcare	5

Education	6
Government / Public Sector	7
Others	8

In which department do you primarily work?	
Finance / Accounting	1
Marketing / Sales	2
Operations / Supply Chain	3
Human Resources	4
Information Technology / Data Analytics	5
Research & Development	6
Other	7

How many years of total professional experience do you have?	
Less than 1 year	1
1–3 years	2
4–7 years	3
8–12 years	4
More than 12 years	5

How would you rate your organization's overall use of business analytics?	
Very Low	
Low	
Moderate	
High	
Very High	

How familiar are you personally with analytics tools (e.g., Power BI, Tableau, Excel, SPSS, Python)?	
Not familiar	1
Slightly familiar	2
Moderately familiar	3

Very familiar	4
Expert level	5

Kindly rate your agreement with the following statements using a Likert scale (1 = Strongly Disagree (SD), 2 = Disagree (D), 3 = Neutral (N), 4 = Agree (A), 5 = Strongly Agree (AG)).						
Statements		**1**	**2**	**3**	**4**	**5**
I.	**Data Quality (DQ)**					
1.	The data available for decision-making in my organization is accurate and reliable.	**1**	**2**	**3**	**4**	**5**
2.	Information we use is complete, without important details missing.	**1**	**2**	**3**	**4**	**5**
3.	Data are updated regularly and available when needed for decisions.	**1**	**2**	**3**	**4**	**5**
4.	The same data give consistent results across different systems or departments.	**1**	**2**	**3**	**4**	**5**
5.	The data we collect are relevant to our strategic and operational needs.	**1**	**2**	**3**	**4**	**5**
II.	Analytics Capability (AC)	**1**	**2**	**3**	**4**	**5**
6.	Our organization has advanced analytical tools that support strategic decisions.	**1**	**2**	**3**	**4**	**5**
7.	Employees have the skills and training needed to analyse and interpret data effectively.	**1**	**2**	**3**	**4**	**5**
8.	Analytics results are integrated into our daily and strategic decision processes.	**1**	**2**	**3**	**4**	**5**
9.	We can quickly transform data into meaningful insights for action.	**1**	**2**	**3**	**4**	**5**
10.	Our analytics systems are flexible enough to handle large and diverse data sources.	**1**	**2**	**3**	**4**	**5**
III.	**Managerial Support (MS)**	**1**	**2**	**3**	**4**	**5**
11.	Top management strongly promotes the use of analytics in decision-making.	**1**	**2**	**3**	**4**	**5**

12.	Senior leaders allocate sufficient financial and technical resources for analytics projects.	**1**	**2**	**3**	**4**	**5**
13.	Managers encourage employees to use data when making decisions.	**1**	**2**	**3**	**4**	**5**
14.	Leadership regularly communicates the importance of evidence-based decisions.	**1**	**2**	**3**	**4**	**5**
15.	Executives act as role models in using analytics for strategic planning.	**1**	**2**	**3**	**4**	**5**
IV.	**Business Analytics Adoption (BAA)**	**1**	**2**	**3**	**4**	**5**
16.	Analytics tools are routinely used across different departments in this organization.	**1**	**2**	**3**	**4**	**5**
17.	Business decisions here rely heavily on insights generated through analytics.	**1**	**2**	**3**	**4**	**5**
18.	Analytics use has become a standard part of our strategic planning process.	**1**	**2**	**3**	**4**	**5**
19.	Our organization continues to invest in new analytics systems and applications.	**1**	**2**	**3**	**4**	**5**
20.	Employees at different levels are encouraged to make use of analytics outputs.	**1**	**2**	**3**	**4**	**5**
V.	**Strategic Decision-Making Effectiveness (SDME)**	**1**	**2**	**3**	**4**	**5**
21.	The use of analytics has improved the accuracy of our strategic decisions.	**1**	**2**	**3**	**4**	**5**
22.	Decisions are made faster because relevant information is readily available.	**1**	**2**	**3**	**4**	**5**
23.	Analytics has helped align decisions more closely with organizational goals.	**1**	**2**	**3**	**4**	**5**
24.	The quality of strategic choices has improved through data-driven insights.	**1**	**2**	**3**	**4**	**5**
25.	Analytics-based decisions have led to better long-term performance outcomes.	**1**	**2**	**3**	**4**	**5**